

Formative Assessment 3

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Case Study: Payroll and Wins in Major League Baseball (1998–2014)

You will investigate whether higher payroll leads to higher win percentage among MLB teams.

You must:

- Tidy the data
- Validate aggregates
- Build regression models
- Perform diagnostic checks
- Interpret results critically

Load necessary libraries

```
library(ggplot2)
library(tidyverse)
```

```
## — Attaching core tidyverse packages — tidyverse 2.0.0 —
## ✓ dplyr      1.2.0    ✓ readr      2.1.6
## ✓ forcats    1.0.1    ✓ stringr    1.6.0
## ✓ lubridate  1.9.5    ✓ tibble     3.3.1
## ✓ purrr      1.2.1    ✓ tidyr      1.3.2
## — Conflicts — tidyverse_conflicts() —
## ✗ dplyr::filter() masks stats::filter()
## ✗ dplyr::lag()     masks stats::lag()
## ⓘ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(ggrepel)
library(broom)
library(cowplot)
```

```
##
## Attaching package: 'cowplot'
##
## The following object is masked from 'package:lubridate':
##
##     stamp
```

```
library(car) # for VIF & diagnostics
```

```
## Loading required package: carData
##
## Attaching package: 'car'
##
## The following object is masked from 'package:dplyr':
##
##     recode
##
## The following object is masked from 'package:purrr':
##
##     some
```

```
library(performance) # optional for model checks
```

Part 1 – Data Wrangling and Validation (6 points)

1.1 Import & Structure (2 points)

- Import as `mlb_raw`.

```
load("C:\\Users\\spike\\Downloads\\ml_pay.rdata")
mlb_raw <- ml_pay
mlb_raw
```

##	payroll	avgwin	Team.name.2014	p1998	p1999	p2000	
## 1	1.1208736	0.4902585	Arizona Diamondbacks	31.61450	70.49600	81.02783	
## 2	1.3817118	0.5527605	Atlanta Braves	61.70800	74.89000	84.53784	
## 3	1.1612117	0.4538250	Baltimore Orioles	71.86092	72.19836	81.44743	
## 4	1.9723587	0.5487172	Boston Red Sox	59.49700	71.72500	77.94033	
## 5	1.4597668	0.4736557	Chicago Cubs	49.81600	42.14276	60.53933	
## 6	1.3153909	0.5111170	Chicago White Sox	35.18000	24.53500	31.13350	
## 7	1.0247816	0.4861602	Cincinnati Reds	20.70733	73.27846	46.86720	
## 8	0.9991810	0.4959225	Cleveland Indians	59.54317	54.44250	75.88087	
## 9	1.0261536	0.4633760	Colorado Rockies	47.71465	55.44350	61.11119	
## 10	1.4297408	0.4822029	Detroit Tigers	19.23750	34.95967	58.26517	
## 11	1.0601501	0.4687202	Houston Astros	48.30400	54.33900	51.28911	
## 12	0.8177417	0.4342288	Kansas City Royals	35.61000	16.55700	23.43300	
## 13	1.5626224	0.5463819	Los Angeles Angels	48.38900	49.86817	51.46417	
## 14	1.7402719	0.5308482	Los Angeles Dodgers	62.80667	71.11579	88.12429	
## 15	0.6678019	0.4813631	Miami Marlins	15.14100	15.15000	20.07200	
## 16	0.9790940	0.4746570	Milwaukee Brewers	31.89790	42.92739	36.50533	
## 17	0.9698272	0.5019047	Minnesota Twins	24.52750	16.34500	16.51950	
## 18	1.5884288	0.4911388	New York Mets	58.66067	71.50643	79.50978	
## 19	2.7032482	0.5830719	New York Yankees	65.66370	88.18071	92.53826	
## 20	0.8409340	0.5445067	Oakland Athletics	22.46350	24.17533	31.97133	
## 21	1.6301209	0.5247021	Philadelphia Phillies	28.62250	30.56817	47.30800	
## 22	0.7339057	0.4371254	Pittsburgh Pirates	13.69500	24.16767	28.92833	
## 23	0.8406668	0.4754884	San Diego Padres	53.06617	45.83218	54.82100	
## 24	1.4168770	0.5304369	San Francisco Giants	48.51472	45.95956	53.73783	
## 25	1.3111203	0.4925819	Seattle Mariners	43.69814	44.39634	58.91500	
## 26	1.3681117	0.5595414	St. Louis Cardinals	44.09085	46.17319	61.45386	
## 27	0.7107894	0.4685176	Tampa Bay Rays	27.37000	38.02750	62.76513	
## 28	1.2693201	0.4956494	Texas Rangers	60.51960	81.57660	70.79592	
## 29	1.1290219	0.4930823	Toronto Blue Jays	34.15850	48.45533	46.23833	
## 30	0.9219641	0.4660195	Washington Nationals	8.31700	16.41300	34.80783	
##	p2001	p2002	p2003	p2004	p2005	p2006	p2007
## 1	81.20651	102.82000	80.64033	70.20498	63.01583	59.68423	52.06755
## 2	91.85169	93.47037	106.24367	88.50779	85.14858	90.15688	87.29083
## 3	72.42633	60.49349	73.87750	51.21265	74.57054	72.58558	93.55481
## 4	109.55891	108.36606	99.94650	125.20854	121.31194	120.09982	143.02621
## 5	64.01583	75.69083	79.86833	91.10167	87.21093	94.42450	99.67033
## 6	62.36300	57.05283	51.01000	65.21250	75.22800	102.75067	108.67183
## 7	45.22788	45.05039	59.35567	43.06786	59.65828	60.90952	68.90498
## 8	91.97498	78.90945	48.58483	34.56930	41.83040	56.03150	61.67327
## 9	71.06800	56.85104	67.17967	64.59040	47.78900	41.23300	54.42400
## 10	49.83117	55.04800	49.16800	46.35355	68.99818	82.61287	95.18037
## 11	60.38267	63.44842	71.04000	74.66630	76.77902	92.55150	87.75900
## 12	35.64300	47.25700	40.51800	47.60900	36.88100	47.29400	67.11650
## 13	46.56818	61.72167	79.03167	101.08467	95.01782	103.47200	109.25133
## 14	108.98095	94.85095	105.87262	89.69434	81.02950	98.44719	108.45452
## 15	35.50417	41.97992	49.05000	42.11804	60.37596	14.99850	30.50700
## 16	43.08933	50.28783	40.62700	27.51850	40.23483	57.56833	70.98650
## 17	24.35000	40.22500	55.50500	53.58500	56.61500	63.39601	71.43950
## 18	93.17443	94.63359	117.17643	100.62930	104.77014	101.08496	115.23166
## 19	109.79189	125.92858	152.74981	182.83551	205.93844	194.66308	189.63905
## 20	33.81075	39.67975	50.26083	59.82517	55.86926	62.24308	79.36694
## 21	41.66417	57.95500	70.78000	93.21917	95.33791	88.27333	89.42821
## 22	52.69833	42.32360	54.81243	32.22793	38.13800	46.71775	38.53783
## 23	38.33312	41.42500	47.92800	54.63950	62.88819	69.89614	58.11057
## 24	63.33267	78.29984	82.85217	82.01917	89.48784	90.05642	90.21900

##	25	75.65250	80.28267	86.95917	81.54383	85.88333	87.95983	106.46083					
##	26	77.27085	74.09827	83.48667	75.63352	93.31984	88.89137	90.28682					
##	27	54.95160	34.38000	19.63000	29.50667	29.89357	35.41797	24.12350					
##	28	88.50442	105.30212	103.49167	54.82597	55.30726	68.22866	68.31867					
##	29	75.79850	76.86433	51.26900	50.01700	45.33650	71.91500	81.94280					
##	30	34.77450	38.67050	51.94850	43.19750	48.58150	63.14300	37.34750					
##		p2008	p2009	p2010	p2011	p2012	p2013	p2014	X2014				
##	1	66.20271	73.57167	60.71817	53.63983	74.28483	89.10050	112.68867	59				
##	2	102.36568	96.72617	84.42367	87.00319	83.30994	89.77819	110.89734	73				
##	3	67.19625	67.10167	81.61250	85.30404	81.42900	90.99333	107.40662	82				
##	4	133.39004	122.69600	162.74733	161.40748	173.18662	150.65550	162.81741	62				
##	5	118.34583	135.05000	146.85900	125.48066	88.19703	104.30468	89.00786	64				
##	6	121.18933	96.06850	108.27320	129.28554	96.91950	119.07328	91.15925	63				
##	7	74.11769	70.96850	72.38654	76.18136	82.20362	107.49130	112.39077	66				
##	8	78.97007	81.62557	61.20397	49.18887	78.43030	77.77280	82.53480	72				
##	9	68.65550	75.20100	84.22700	87.99807	78.06957	71.92407	95.83207	57				
##	10	137.68520	115.08514	122.86493	105.70523	132.30000	148.41450	162.22853	77				
##	11	88.93041	102.99641	92.35550	70.69400	60.65100	22.06260	44.54417	62				
##	12	58.24550	70.90833	72.26771	36.12640	60.91622	81.49173	92.03435	78				
##	13	119.21633	113.70900	105.01367	138.99852	154.48517	127.89625	155.69200	85				
##	14	118.58854	100.45810	94.94552	103.78899	95.14357	216.59758	235.29522	79				
##	15	21.81150	36.81400	55.64150	56.94400	118.07800	36.34190	47.56540	68				
##	16	80.93750	80.25750	81.10828	85.49733	97.65394	82.97694	103.84481	74				
##	17	56.93277	65.29927	97.55917	112.73700	94.08500	75.80250	85.77650	61				
##	18	137.79338	135.77399	132.70145	120.14731	93.35398	73.39665	89.05176	67				
##	19	209.08158	201.44929	206.33339	201.68903	197.96229	228.83549	203.81251	72				
##	20	47.96713	62.31000	51.65490	66.53650	55.37250	60.66450	83.40140	79				
##	21	98.26988	113.00405	141.92738	172.97638	174.53894	165.38571	180.05272	65				
##	22	48.68978	48.74300	34.94300	46.04700	63.43200	79.55500	78.11167	71				
##	23	73.67762	42.79670	37.79930	45.86914	55.24470	67.14360	90.09420	66				
##	24	76.59450	82.16145	97.82883	118.21633	117.62068	140.26433	154.18588	77				
##	25	117.66648	98.90417	98.37667	86.42460	81.97810	72.03114	92.08194	77				
##	26	99.62445	88.52841	93.54075	105.43357	110.30086	115.22209	111.02036	77				
##	27	43.82060	63.31303	71.92347	41.93217	64.17350	57.89527	77.06289	68				
##	28	67.71233	68.64602	55.25055	92.29927	120.51097	114.09010	136.03617	53				
##	29	97.79390	80.99366	62.68936	62.51780	75.48920	117.52780	132.62870	72				
##	30	54.96100	59.32800	61.42500	63.68193	81.33614	114.05677	134.70444	79				
##		X2013	X2012	X2011	X2010	X2009	X2008	X2007	X2006	X2005	X2004	X2003	X2002
##	1	81	81	94	65	70	82	90	76	77	51	84	98
##	2	96	94	89	91	86	72	84	79	90	96	101	101
##	3	85	93	69	66	64	68	69	70	74	78	71	67
##	4	97	69	90	89	95	95	96	86	95	98	95	93
##	5	66	61	71	75	83	97	85	66	79	89	88	67
##	6	63	85	79	88	79	89	72	90	99	83	86	81
##	7	90	97	79	91	78	74	72	80	73	76	69	78
##	8	92	68	80	69	65	81	96	78	93	80	68	74
##	9	74	64	73	83	92	74	90	76	67	68	74	73
##	10	93	88	95	81	86	74	88	95	71	72	43	55
##	11	51	55	56	76	74	86	73	82	89	92	87	84
##	12	86	72	71	67	65	75	69	62	56	58	83	62
##	13	78	89	86	80	97	100	94	89	95	92	77	99
##	14	92	86	82	80	95	84	82	88	71	93	85	92
##	15	62	69	72	80	87	84	71	78	83	83	91	79
##	16	74	83	96	77	80	90	83	75	81	67	68	56
##	17	66	66	63	94	87	88	79	96	83	92	90	94
##	18	74	74	77	79	70	89	88	97	83	71	66	75

## 19	85	95	97	95	103	89	94	97	95	101	101	103
## 20	96	94	74	81	75	75	76	93	88	91	96	103
## 21	73	81	102	97	93	92	89	85	88	86	86	80
## 22	94	79	72	57	62	67	68	67	67	72	75	72
## 23	76	76	71	90	75	63	89	88	82	87	64	66
## 24	76	94	86	92	88	72	71	76	75	91	100	95
## 25	71	75	67	61	85	61	88	78	69	63	93	93
## 26	97	88	90	86	91	86	78	83	100	105	85	97
## 27	92	90	91	96	84	97	66	61	67	70	63	55
## 28	91	93	96	90	87	79	75	80	79	89	71	72
## 29	74	73	81	85	75	86	83	87	80	67	86	78
## 30	86	98	80	69	59	59	73	71	81	67	83	83
##	X2001	X2000	X1999	X1998	X2014.pct	X2013.pct	X2012.pct	X2011.pct	X2010.pct			
## 1	92	85	100	65	0.4154930	0.4969325	0.5000000	0.5802469	0.4012346			
## 2	88	95	103	106	0.5140845	0.5889571	0.5802469	0.5493827	0.5617284			
## 3	63	74	78	79	0.5774648	0.5214724	0.5740741	0.4259259	0.4074074			
## 4	82	85	94	92	0.4366197	0.5950920	0.4259259	0.5555556	0.5493827			
## 5	88	65	67	90	0.4507042	0.4049080	0.3765432	0.4382716	0.4629630			
## 6	83	95	75	80	0.4436620	0.3865031	0.5246914	0.4876543	0.5432099			
## 7	66	85	96	77	0.4647887	0.5521472	0.5987654	0.4876543	0.5617284			
## 8	91	90	97	89	0.5070423	0.5644172	0.4197531	0.4938272	0.4259259			
## 9	73	82	72	77	0.4014085	0.4539877	0.3950617	0.4506173	0.5123457			
## 10	66	79	69	65	0.5422535	0.5705521	0.5432099	0.5864198	0.5000000			
## 11	93	72	97	102	0.4366197	0.3128834	0.3395062	0.3456790	0.4691358			
## 12	65	77	64	72	0.5492958	0.5276074	0.4444444	0.4382716	0.4135802			
## 13	75	82	70	85	0.5985915	0.4785276	0.5493827	0.5308642	0.4938272			
## 14	86	86	77	83	0.5563380	0.5644172	0.5308642	0.5061728	0.4938272			
## 15	76	79	64	54	0.4788732	0.3803681	0.4259259	0.4444444	0.4938272			
## 16	68	73	74	74	0.5211268	0.4539877	0.5123457	0.5925926	0.4753086			
## 17	85	69	63	70	0.4295775	0.4049080	0.4074074	0.3888889	0.5802469			
## 18	82	94	97	88	0.4718310	0.4539877	0.4567901	0.4753086	0.4876543			
## 19	95	87	98	114	0.5070423	0.5214724	0.5864198	0.5987654	0.5864198			
## 20	102	91	87	74	0.5563380	0.5889571	0.5802469	0.4567901	0.5000000			
## 21	86	65	77	75	0.4577465	0.4478528	0.5000000	0.6296296	0.5987654			
## 22	62	69	78	69	0.5000000	0.5766871	0.4876543	0.4444444	0.3518519			
## 23	79	76	74	98	0.4647887	0.4662577	0.4691358	0.4382716	0.5555556			
## 24	90	97	86	89	0.5422535	0.4662577	0.5802469	0.5308642	0.5679012			
## 25	116	91	79	76	0.5422535	0.4355828	0.4629630	0.4135802	0.3765432			
## 26	93	95	75	83	0.5422535	0.5950920	0.5432099	0.5555556	0.5308642			
## 27	62	69	69	63	0.4788732	0.5644172	0.5555556	0.5617284	0.5925926			
## 28	73	71	95	88	0.3732394	0.5582822	0.5740741	0.5925926	0.5555556			
## 29	80	83	84	88	0.5070423	0.4539877	0.4506173	0.5000000	0.5246914			
## 30	68	67	68	65	0.5563380	0.5276074	0.6049383	0.4938272	0.4259259			
##	X2009.pct	X2008.pct	X2007.pct	X2006.pct	X2005.pct	X2004.pct	X2003.pct					
## 1	0.4294479	0.5030675	0.5521472	0.4691358	0.4753086	0.3148148	0.5185185					
## 2	0.5276074	0.4417178	0.5153374	0.4876543	0.5555556	0.5925926	0.6234568					
## 3	0.3926380	0.4171779	0.4233129	0.4320988	0.4567901	0.4814815	0.4382716					
## 4	0.5828221	0.5828221	0.5889571	0.5308642	0.5864198	0.6049383	0.5864198					
## 5	0.5092025	0.5950920	0.5214724	0.4074074	0.4876543	0.5493827	0.5432099					
## 6	0.4846626	0.5460123	0.4417178	0.5555556	0.6111111	0.5123457	0.5308642					
## 7	0.4785276	0.4539877	0.4417178	0.4938272	0.4506173	0.4691358	0.4259259					
## 8	0.3987730	0.4969325	0.5889571	0.4814815	0.5740741	0.4938272	0.4197531					
## 9	0.5644172	0.4539877	0.5521472	0.4691358	0.4135802	0.4197531	0.4567901					
## 10	0.5276074	0.4539877	0.5398773	0.5864198	0.4382716	0.4444444	0.2654321					
## 11	0.4539877	0.5276074	0.4478528	0.5061728	0.5493827	0.5679012	0.5370370					
## 12	0.3987730	0.4601227	0.4233129	0.3827160	0.3456790	0.3580247	0.5123457					

```

## 13 0.5950920 0.6134969 0.5766871 0.5493827 0.5864198 0.5679012 0.4753086
## 14 0.5828221 0.5153374 0.5030675 0.5432099 0.4382716 0.5740741 0.5246914
## 15 0.5337423 0.5153374 0.4355828 0.4814815 0.5123457 0.5123457 0.5617284
## 16 0.4907975 0.5521472 0.5092025 0.4629630 0.5000000 0.4135802 0.4197531
## 17 0.5337423 0.5398773 0.4846626 0.5925926 0.5123457 0.5679012 0.5555556
## 18 0.4294479 0.5460123 0.5398773 0.5987654 0.5123457 0.4382716 0.4074074
## 19 0.6319018 0.5460123 0.5766871 0.5987654 0.5864198 0.6234568 0.6234568
## 20 0.4601227 0.4601227 0.4662577 0.5740741 0.5432099 0.5617284 0.5925926
## 21 0.5705521 0.5644172 0.5460123 0.5246914 0.5432099 0.5308642 0.5308642
## 22 0.3803681 0.4110429 0.4171779 0.4135802 0.4135802 0.4444444 0.4629630
## 23 0.4601227 0.3865031 0.5460123 0.5432099 0.5061728 0.5370370 0.3950617
## 24 0.5398773 0.4417178 0.4355828 0.4691358 0.4629630 0.5617284 0.6172840
## 25 0.5214724 0.3742331 0.5398773 0.4814815 0.4259259 0.3888889 0.5740741
## 26 0.5582822 0.5276074 0.4785276 0.5123457 0.6172840 0.6481481 0.5246914
## 27 0.5153374 0.5950920 0.4049080 0.3765432 0.4135802 0.4320988 0.3888889
## 28 0.5337423 0.4846626 0.4601227 0.4938272 0.4876543 0.5493827 0.4382716
## 29 0.4601227 0.5276074 0.5092025 0.5370370 0.4938272 0.4135802 0.5308642
## 30 0.3619632 0.3619632 0.4478528 0.4382716 0.5000000 0.4135802 0.5123457
##      X2002.pct X2001.pct X2000.pct X1999.pct X1998.pct
## 1 0.6049383 0.5679012 0.5246914 0.6134969 0.3987730
## 2 0.6234568 0.5432099 0.5864198 0.6319018 0.6503067
## 3 0.4135802 0.3888889 0.4567901 0.4785276 0.4846626
## 4 0.5740741 0.5061728 0.5246914 0.5766871 0.5644172
## 5 0.4135802 0.5432099 0.4012346 0.4110429 0.5521472
## 6 0.5000000 0.5123457 0.5864198 0.4601227 0.4907975
## 7 0.4814815 0.4074074 0.5246914 0.5889571 0.4723926
## 8 0.4567901 0.5617284 0.5555556 0.5950920 0.5460123
## 9 0.4506173 0.4506173 0.5061728 0.4417178 0.4723926
## 10 0.3395062 0.4074074 0.4876543 0.4233129 0.3987730
## 11 0.5185185 0.5740741 0.4444444 0.5950920 0.6257669
## 12 0.3827160 0.4012346 0.4753086 0.3926380 0.4417178
## 13 0.6111111 0.4629630 0.5061728 0.4294479 0.5214724
## 14 0.5679012 0.5308642 0.5308642 0.4723926 0.5092025
## 15 0.4876543 0.4691358 0.4876543 0.3926380 0.3312883
## 16 0.3456790 0.4197531 0.4506173 0.4539877 0.4539877
## 17 0.5802469 0.5246914 0.4259259 0.3865031 0.4294479
## 18 0.4629630 0.5061728 0.5802469 0.5950920 0.5398773
## 19 0.6358025 0.5864198 0.5370370 0.6012270 0.6993865
## 20 0.6358025 0.6296296 0.5617284 0.5337423 0.4539877
## 21 0.4938272 0.5308642 0.4012346 0.4723926 0.4601227
## 22 0.4444444 0.3827160 0.4259259 0.4785276 0.4233129
## 23 0.4074074 0.4876543 0.4691358 0.4539877 0.6012270
## 24 0.5864198 0.5555556 0.5987654 0.5276074 0.5460123
## 25 0.5740741 0.7160494 0.5617284 0.4846626 0.4662577
## 26 0.5987654 0.5740741 0.5864198 0.4601227 0.5092025
## 27 0.3395062 0.3827160 0.4259259 0.4233129 0.3865031
## 28 0.4444444 0.4506173 0.4382716 0.5828221 0.5398773
## 29 0.4814815 0.4938272 0.5123457 0.5153374 0.5398773
## 30 0.5123457 0.4197531 0.4135802 0.4171779 0.3987730

```

```
colnames(mlb_raw)
```

```
## [1] "payroll"      "avgwin"      "Team.name.2014" "p1998"
## [5] "p1999"      "p2000"      "p2001"      "p2002"
## [9] "p2003"      "p2004"      "p2005"      "p2006"
## [13] "p2007"      "p2008"      "p2009"      "p2010"
## [17] "p2011"      "p2012"      "p2013"      "p2014"
## [21] "X2014"      "X2013"      "X2012"      "X2011"
## [25] "X2010"      "X2009"      "X2008"      "X2007"
## [29] "X2006"      "X2005"      "X2004"      "X2003"
## [33] "X2002"      "X2001"      "X2000"      "X1999"
## [37] "X1998"      "X2014.pct"  "X2013.pct"  "X2012.pct"
## [41] "X2011.pct"  "X2010.pct"  "X2009.pct"  "X2008.pct"
## [45] "X2007.pct"  "X2006.pct"  "X2005.pct"  "X2004.pct"
## [49] "X2003.pct"  "X2002.pct"  "X2001.pct"  "X2000.pct"
## [53] "X1999.pct"  "X1998.pct"
```

- Report dimensions and variable types.

```
dim(mlb_raw)
```

```
## [1] 30 54
```

```
str(mlb_raw)
```

```

## 'data.frame':      30 obs. of  54 variables:
## $ payroll      : num  1.12 1.38 1.16 1.97 1.46 ...
## $ avgwin       : num  0.49 0.553 0.454 0.549 0.474 ...
## $ Team.name.2014: Factor w/ 30 levels "Arizona Diamondbacks",...: 1 2 3 4 5 6 7 8 9 10 ...
## $ p1998        : num  31.6 61.7 71.9 59.5 49.8 ...
## $ p1999        : num  70.5 74.9 72.2 71.7 42.1 ...
## $ p2000        : num  81 84.5 81.4 77.9 60.5 ...
## $ p2001        : num  81.2 91.9 72.4 109.6 64 ...
## $ p2002        : num  102.8 93.5 60.5 108.4 75.7 ...
## $ p2003        : num  80.6 106.2 73.9 99.9 79.9 ...
## $ p2004        : num  70.2 88.5 51.2 125.2 91.1 ...
## $ p2005        : num  63 85.1 74.6 121.3 87.2 ...
## $ p2006        : num  59.7 90.2 72.6 120.1 94.4 ...
## $ p2007        : num  52.1 87.3 93.6 143 99.7 ...
## $ p2008        : num  66.2 102.4 67.2 133.4 118.3 ...
## $ p2009        : num  73.6 96.7 67.1 122.7 135.1 ...
## $ p2010        : num  60.7 84.4 81.6 162.7 146.9 ...
## $ p2011        : num  53.6 87 85.3 161.4 125.5 ...
## $ p2012        : num  74.3 83.3 81.4 173.2 88.2 ...
## $ p2013        : num  89.1 89.8 91 150.7 104.3 ...
## $ p2014        : num  113 111 107 163 89 ...
## $ X2014        : int  59 73 82 62 64 63 66 72 57 77 ...
## $ X2013        : int  81 96 85 97 66 63 90 92 74 93 ...
## $ X2012        : int  81 94 93 69 61 85 97 68 64 88 ...
## $ X2011        : int  94 89 69 90 71 79 79 80 73 95 ...
## $ X2010        : int  65 91 66 89 75 88 91 69 83 81 ...
## $ X2009        : int  70 86 64 95 83 79 78 65 92 86 ...
## $ X2008        : int  82 72 68 95 97 89 74 81 74 74 ...
## $ X2007        : int  90 84 69 96 85 72 72 96 90 88 ...
## $ X2006        : int  76 79 70 86 66 90 80 78 76 95 ...
## $ X2005        : int  77 90 74 95 79 99 73 93 67 71 ...
## $ X2004        : int  51 96 78 98 89 83 76 80 68 72 ...
## $ X2003        : int  84 101 71 95 88 86 69 68 74 43 ...
## $ X2002        : int  98 101 67 93 67 81 78 74 73 55 ...
## $ X2001        : int  92 88 63 82 88 83 66 91 73 66 ...
## $ X2000        : int  85 95 74 85 65 95 85 90 82 79 ...
## $ X1999        : int  100 103 78 94 67 75 96 97 72 69 ...
## $ X1998        : int  65 106 79 92 90 80 77 89 77 65 ...
## $ X2014.pct     : num  0.415 0.514 0.577 0.437 0.451 ...
## $ X2013.pct     : num  0.497 0.589 0.521 0.595 0.405 ...
## $ X2012.pct     : num  0.5 0.58 0.574 0.426 0.377 ...
## $ X2011.pct     : num  0.58 0.549 0.426 0.556 0.438 ...
## $ X2010.pct     : num  0.401 0.562 0.407 0.549 0.463 ...
## $ X2009.pct     : num  0.429 0.528 0.393 0.583 0.509 ...
## $ X2008.pct     : num  0.503 0.442 0.417 0.583 0.595 ...
## $ X2007.pct     : num  0.552 0.515 0.423 0.589 0.521 ...
## $ X2006.pct     : num  0.469 0.488 0.432 0.531 0.407 ...
## $ X2005.pct     : num  0.475 0.556 0.457 0.586 0.488 ...
## $ X2004.pct     : num  0.315 0.593 0.481 0.605 0.549 ...
## $ X2003.pct     : num  0.519 0.623 0.438 0.586 0.543 ...
## $ X2002.pct     : num  0.605 0.623 0.414 0.574 0.414 ...
## $ X2001.pct     : num  0.568 0.543 0.389 0.506 0.543 ...
## $ X2000.pct     : num  0.525 0.586 0.457 0.525 0.401 ...
## $ X1999.pct     : num  0.613 0.632 0.479 0.577 0.411 ...
## $ X1998.pct     : num  0.399 0.65 0.485 0.564 0.552 ...

```



```
glimpse(mlb_raw)
```

```

## Rows: 30
## Columns: 54
## $ payroll      <dbl> 1.1208736, 1.3817118, 1.1612117, 1.9723587, 1.4597668, ...
## $ avgwin       <dbl> 0.4902585, 0.5527605, 0.4538250, 0.5487172, 0.4736557, ...
## $ Team.name.2014 <fct> Arizona Diamondbacks, Atlanta Braves, Baltimore Orioles...
## $ p1998        <dbl> 31.61450, 61.70800, 71.86092, 59.49700, 49.81600, 35.18...
## $ p1999        <dbl> 70.49600, 74.89000, 72.19836, 71.72500, 42.14276, 24.53...
## $ p2000        <dbl> 81.02783, 84.53784, 81.44743, 77.94033, 60.53933, 31.13...
## $ p2001        <dbl> 81.20651, 91.85169, 72.42633, 109.55891, 64.01583, 62.3...
## $ p2002        <dbl> 102.82000, 93.47037, 60.49349, 108.36606, 75.69083, 57....
## $ p2003        <dbl> 80.64033, 106.24367, 73.87750, 99.94650, 79.86833, 51.0...
## $ p2004        <dbl> 70.20498, 88.50779, 51.21265, 125.20854, 91.10167, 65.2...
## $ p2005        <dbl> 63.01583, 85.14858, 74.57054, 121.31194, 87.21093, 75.2...
## $ p2006        <dbl> 59.68423, 90.15688, 72.58558, 120.09982, 94.42450, 102....
## $ p2007        <dbl> 52.06755, 87.29083, 93.55481, 143.02621, 99.67033, 108....
## $ p2008        <dbl> 66.20271, 102.36568, 67.19625, 133.39004, 118.34583, 12...
## $ p2009        <dbl> 73.57167, 96.72617, 67.10167, 122.69600, 135.05000, 96...
## $ p2010        <dbl> 60.71817, 84.42367, 81.61250, 162.74733, 146.85900, 108...
## $ p2011        <dbl> 53.63983, 87.00319, 85.30404, 161.40748, 125.48066, 129...
## $ p2012        <dbl> 74.28483, 83.30994, 81.42900, 173.18662, 88.19703, 96.9...
## $ p2013        <dbl> 89.10050, 89.77819, 90.99333, 150.65550, 104.30468, 119...
## $ p2014        <dbl> 112.68867, 110.89734, 107.40662, 162.81741, 89.00786, 9...
## $ X2014        <int> 59, 73, 82, 62, 64, 63, 66, 72, 57, 77, 62, 78, 85, 79,...
## $ X2013        <int> 81, 96, 85, 97, 66, 63, 90, 92, 74, 93, 51, 86, 78, 92,...
## $ X2012        <int> 81, 94, 93, 69, 61, 85, 97, 68, 64, 88, 55, 72, 89, 86,...
## $ X2011        <int> 94, 89, 69, 90, 71, 79, 79, 80, 73, 95, 56, 71, 86, 82,...
## $ X2010        <int> 65, 91, 66, 89, 75, 88, 91, 69, 83, 81, 76, 67, 80, 80,...
## $ X2009        <int> 70, 86, 64, 95, 83, 79, 78, 65, 92, 86, 74, 65, 97, 95,...
## $ X2008        <int> 82, 72, 68, 95, 97, 89, 74, 81, 74, 74, 86, 75, 100, 84...
## $ X2007        <int> 90, 84, 69, 96, 85, 72, 72, 96, 90, 88, 73, 69, 94, 82,...
## $ X2006        <int> 76, 79, 70, 86, 66, 90, 80, 78, 76, 95, 82, 62, 89, 88,...
## $ X2005        <int> 77, 90, 74, 95, 79, 99, 73, 93, 67, 71, 89, 56, 95, 71,...
## $ X2004        <int> 51, 96, 78, 98, 89, 83, 76, 80, 68, 72, 92, 58, 92, 93,...
## $ X2003        <int> 84, 101, 71, 95, 88, 86, 69, 68, 74, 43, 87, 83, 77, 85...
## $ X2002        <int> 98, 101, 67, 93, 67, 81, 78, 74, 73, 55, 84, 62, 99, 92...
## $ X2001        <int> 92, 88, 63, 82, 88, 83, 66, 91, 73, 66, 93, 65, 75, 86,...
## $ X2000        <int> 85, 95, 74, 85, 65, 95, 85, 90, 82, 79, 72, 77, 82, 86,...
## $ X1999        <int> 100, 103, 78, 94, 67, 75, 96, 97, 72, 69, 97, 64, 70, 7...
## $ X1998        <int> 65, 106, 79, 92, 90, 80, 77, 89, 77, 65, 102, 72, 85, 8...
## $ X2014.pct    <dbl> 0.4154930, 0.5140845, 0.5774648, 0.4366197, 0.4507042, ...
## $ X2013.pct    <dbl> 0.4969325, 0.5889571, 0.5214724, 0.5950920, 0.4049080, ...
## $ X2012.pct    <dbl> 0.5000000, 0.5802469, 0.5740741, 0.4259259, 0.3765432, ...
## $ X2011.pct    <dbl> 0.5802469, 0.5493827, 0.4259259, 0.5555556, 0.4382716, ...
## $ X2010.pct    <dbl> 0.4012346, 0.5617284, 0.4074074, 0.5493827, 0.4629630, ...
## $ X2009.pct    <dbl> 0.4294479, 0.5276074, 0.3926380, 0.5828221, 0.5092025, ...
## $ X2008.pct    <dbl> 0.5030675, 0.4417178, 0.4171779, 0.5828221, 0.5950920, ...
## $ X2007.pct    <dbl> 0.5521472, 0.5153374, 0.4233129, 0.5889571, 0.5214724, ...
## $ X2006.pct    <dbl> 0.4691358, 0.4876543, 0.4320988, 0.5308642, 0.4074074, ...
## $ X2005.pct    <dbl> 0.4753086, 0.5555556, 0.4567901, 0.5864198, 0.4876543, ...
## $ X2004.pct    <dbl> 0.3148148, 0.5925926, 0.4814815, 0.6049383, 0.5493827, ...
## $ X2003.pct    <dbl> 0.5185185, 0.6234568, 0.4382716, 0.5864198, 0.5432099, ...
## $ X2002.pct    <dbl> 0.6049383, 0.6234568, 0.4135802, 0.5740741, 0.4135802, ...
## $ X2001.pct    <dbl> 0.5679012, 0.5432099, 0.3888889, 0.5061728, 0.5432099, ...
## $ X2000.pct    <dbl> 0.5246914, 0.5864198, 0.4567901, 0.5246914, 0.4012346, ...

```

```
## $ X1999.pct      <dbl> 0.6134969, 0.6319018, 0.4785276, 0.5766871, 0.4110429, ...  
## $ X1998.pct      <dbl> 0.3987730, 0.6503067, 0.4846626, 0.5644172, 0.5521472, ...
```

- Briefly comment on structure.

The dataset contains 30 observations (rows) representing Major League Baseball teams and 54 variables (columns). The data are currently structured in a wide format, where each team occupies one row and yearly information (1998–2014) is spread across multiple columns.

Payroll, wins, and win percentage for each year are stored as separate variables (e.g., p1998, X1998, X1998.pct).

This structure is not ideal for regression analysis because it does not follow tidy data principles. To properly analyze payroll and win percentage relationships, the dataset must be reshaped into a long (panel) format, where each row represents a team year observation.

1.2 Tidy Data (4 points)

`mlb_yearly`

- team
- year
- payroll
- pct_wins
- num_wins

Rename team column first:

```
mlb_clean <- mlb_raw %>%  
  rename(team = Team.name.2014)
```

Pivot payroll columns (p1998-p2014)

```
payroll_long <- mlb_clean %>%  
  select(team, matches("^p[0-9]{4}$")) %>%  
  pivot_longer(  
    cols = -team,  
    names_to = "year",  
    values_to = "payroll"  
  ) %>%  
  mutate(year = as.numeric(str_remove(year, "p")))
```

Pivot wins columns (x1998-x2014)

```
wins_long <- mlb_clean %>%  
  select(team, starts_with("X"), -contains("pct")) %>%  
  pivot_longer(  
    cols = starts_with("X"),  
    names_to = "year",  
    values_to = "num_wins"  
  ) %>%  
  mutate(year = as.numeric(str_remove(year, "X")))
```

Pivot win percentage columns (X1998.pct–X2014.pct)

```
pct_long <- mlb_clean %>%
  select(team, contains(".pct")) %>%
  pivot_longer(
    cols = contains(".pct"),
    names_to = "year",
    values_to = "pct_wins"
  ) %>%
  mutate(year = as.numeric(str_remove_all(year, "X|\\.pct")))
```

Merge everything together

```
mlb_yearly <- payroll_long %>%
  left_join(wins_long, by = c("team", "year")) %>%
  left_join(pct_long, by = c("team", "year")) %>%
  arrange(team, year)
```

```
dim(mlb_yearly)
```

```
## [1] 510    5
```

```
colnames(mlb_yearly)
```

```
## [1] "team"      "year"      "payroll"   "num_wins" "pct_wins"
```

```
mlb_yearly
```

```
## # A tibble: 510 × 5
##   team          year payroll num_wins pct_wins
##   <fct>        <dbl>   <dbl>   <int>   <dbl>
## 1 Arizona Diamondbacks 1998    31.6     65    0.399
## 2 Arizona Diamondbacks 1999    70.5    100    0.613
## 3 Arizona Diamondbacks 2000    81.0     85    0.525
## 4 Arizona Diamondbacks 2001    81.2     92    0.568
## 5 Arizona Diamondbacks 2002   103.     98    0.605
## 6 Arizona Diamondbacks 2003    80.6     84    0.519
## 7 Arizona Diamondbacks 2004    70.2     51    0.315
## 8 Arizona Diamondbacks 2005    63.0     77    0.475
## 9 Arizona Diamondbacks 2006    59.7     76    0.469
## 10 Arizona Diamondbacks 2007    52.1     90    0.552
## # i 500 more rows
```

Create:

mlb_aggregate

- team
- payroll_aggregate
- pct_wins_aggregate

```
mlb_aggregate <- mlb_yearly %>%
  group_by(team) %>%
  summarise(
    payroll_aggregate = sum(payroll, na.rm = TRUE),
    pct_wins_aggregate = mean(pct_wins, na.rm = TRUE),
    .groups = "drop"
  )
```

```
dim(mlb_aggregate)
```

```
## [1] 30  3
```

```
colnames(mlb_aggregate)
```

```
## [1] "team"          "payroll_aggregate" "pct_wins_aggregate"
```

```
mlb_aggregate
```

```
## # A tibble: 30 × 3
##   team                payroll_aggregate pct_wins_aggregate
##   <fct>                <dbl>                <dbl>
## 1 Arizona Diamondbacks      1223.                0.492
## 2 Atlanta Braves            1518.                0.563
## 3 Baltimore Orioles         1305.                0.457
## 4 Boston Red Sox             2104.                0.551
## 5 Chicago Cubs               1552.                0.475
## 6 Chicago White Sox          1375.                0.507
## 7 Cincinnati Reds           1119.                0.491
## 8 Cleveland Indians          1113.                0.505
## 9 Colorado Rockies           1129.                0.463
## 10 Detroit Tigers            1484.                0.474
## # i 20 more rows
```

The original data set was in wide format, where yearly payroll, wins, and win percentages were stored as separate columns. The data were reshaped into long format to satisfy tidy data principles, such that each row represents a single team–year observation.

Since the data set includes 30 MLB teams and covers 17 seasons (1998–2014), the expected number of observations is:

$30 \times 17 = 510$.

The resulting `mlb_yearly` dataset contains 510 rows and 5 variables (`team`, `year`, `payroll`, `pct_wins`, `num_wins`), confirming a balanced panel data set appropriate for regression analysis.

The `mlb_aggregate` data set summarizes each team’s total payroll and average win percentage across all seasons.

Part 2 – Aggregate Consistency Check (4 points)

1. Compute aggregates from `mlb_yearly`.

```
mlb_computed <- mlb_yearly %>%
  group_by(team) %>%
  summarise(
    payroll_computed = sum(payroll, na.rm = TRUE),
    pct_wins_computed = mean(pct_wins, na.rm = TRUE),
    .groups = "drop"
  )
```

mlb_computed

```
## # A tibble: 30 × 3
##   team                payroll_computed pct_wins_computed
##   <fct>                <dbl>                <dbl>
## 1 Arizona Diamondbacks      1223.                0.492
## 2 Atlanta Braves            1518.                0.563
## 3 Baltimore Orioles         1305.                0.457
## 4 Boston Red Sox            2104.                0.551
## 5 Chicago Cubs              1552.                0.475
## 6 Chicago White Sox         1375.                0.507
## 7 Cincinnati Reds          1119.                0.491
## 8 Cleveland Indians         1113.                0.505
## 9 Colorado Rockies          1129.                0.463
## 10 Detroit Tigers           1484.                0.474
## # i 20 more rows
```

2. Join with provided aggregates.

```
mlb_provided <- mlb_raw %>%
  rename(team = Team.name.2014) %>%
  select(team,
    payroll_provided = payroll,
    pct_wins_provided = avgwin)
```

mlb_provided

##	team	payroll_provided	pct_wins_provided
## 1	Arizona Diamondbacks	1.1208736	0.4902585
## 2	Atlanta Braves	1.3817118	0.5527605
## 3	Baltimore Orioles	1.1612117	0.4538250
## 4	Boston Red Sox	1.9723587	0.5487172
## 5	Chicago Cubs	1.4597668	0.4736557
## 6	Chicago White Sox	1.3153909	0.5111170
## 7	Cincinnati Reds	1.0247816	0.4861602
## 8	Cleveland Indians	0.9991810	0.4959225
## 9	Colorado Rockies	1.0261536	0.4633760
## 10	Detroit Tigers	1.4297408	0.4822029
## 11	Houston Astros	1.0601501	0.4687202
## 12	Kansas City Royals	0.8177417	0.4342288
## 13	Los Angeles Angels	1.5626224	0.5463819
## 14	Los Angeles Dodgers	1.7402719	0.5308482
## 15	Miami Marlins	0.6678019	0.4813631
## 16	Milwaukee Brewers	0.9790940	0.4746570
## 17	Minnesota Twins	0.9698272	0.5019047
## 18	New York Mets	1.5884288	0.4911388
## 19	New York Yankees	2.7032482	0.5830719
## 20	Oakland Athletics	0.8409340	0.5445067
## 21	Philadelphia Phillies	1.6301209	0.5247021
## 22	Pittsburgh Pirates	0.7339057	0.4371254
## 23	San Diego Padres	0.8406668	0.4754884
## 24	San Francisco Giants	1.4168770	0.5304369
## 25	Seattle Mariners	1.3111203	0.4925819
## 26	St. Louis Cardinals	1.3681117	0.5595414
## 27	Tampa Bay Rays	0.7107894	0.4685176
## 28	Texas Rangers	1.2693201	0.4956494
## 29	Toronto Blue Jays	1.1290219	0.4930823
## 30	Washington Nationals	0.9219641	0.4660195

Join Computed and Provided Aggregates

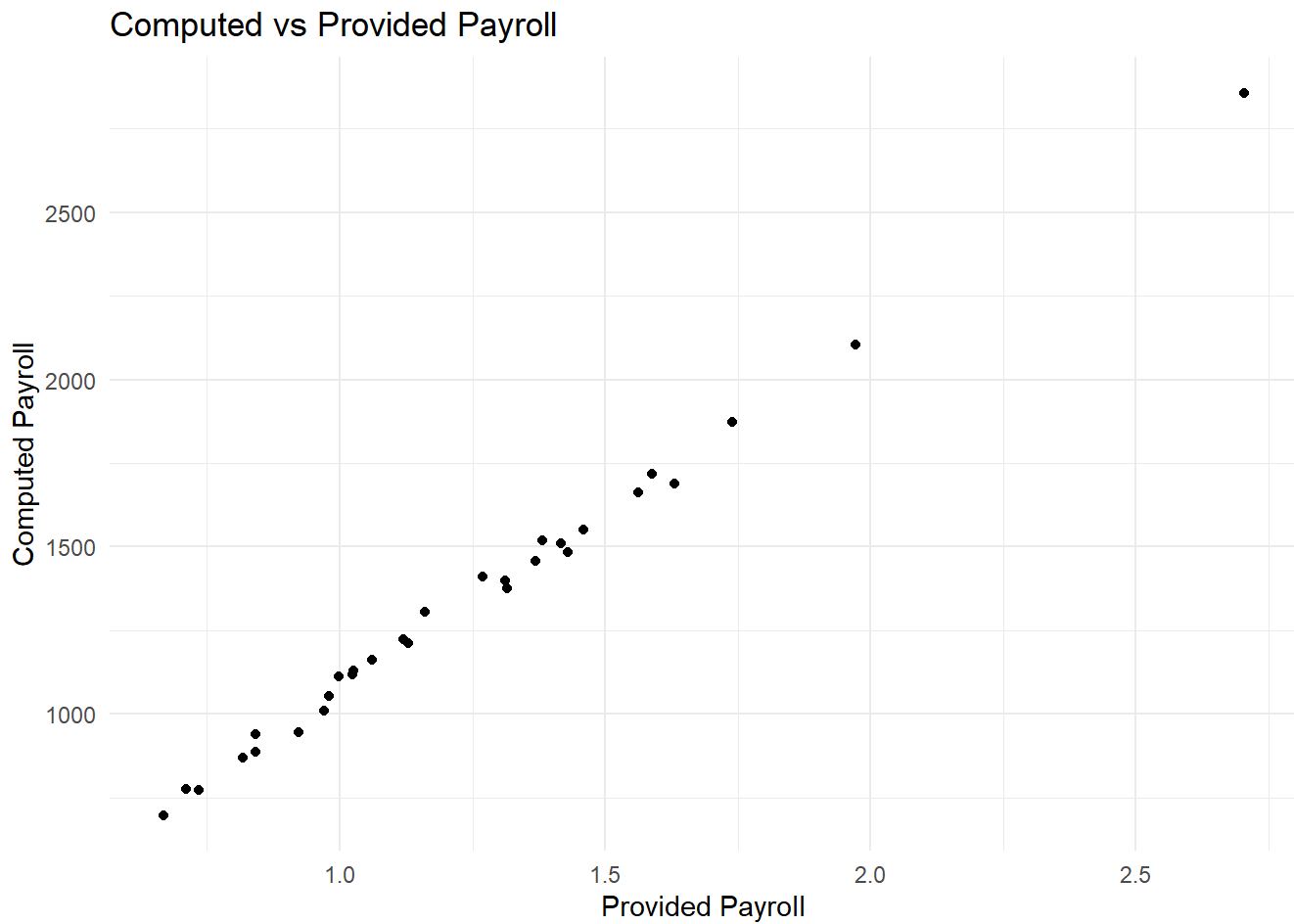
```
mlb_check <- mlb_computed %>%
  left_join(mlb_provided, by = "team")
```

```
mlb_check
```

```
## # A tibble: 30 × 5
##   team      payroll_computed pct_wins_computed payroll_provided pct_wins_provided
##   <fct>          <dbl>          <dbl>          <dbl>          <dbl>
## 1 Arizon...      1223.          0.492          1.12          0.490
## 2 Atlant...     1518.          0.563          1.38          0.553
## 3 Baltim...     1305.          0.457          1.16          0.454
## 4 Boston...     2104.          0.551          1.97          0.549
## 5 Chicag...     1552.          0.475          1.46          0.474
## 6 Chicag...     1375.          0.507          1.32          0.511
## 7 Cincin...     1119.          0.491          1.02          0.486
## 8 Clevel...     1113.          0.505          0.999          0.496
## 9 Colora...     1129.          0.463          1.03          0.463
## 10 Detroi...     1484.          0.474          1.43          0.482
## # i 20 more rows
```

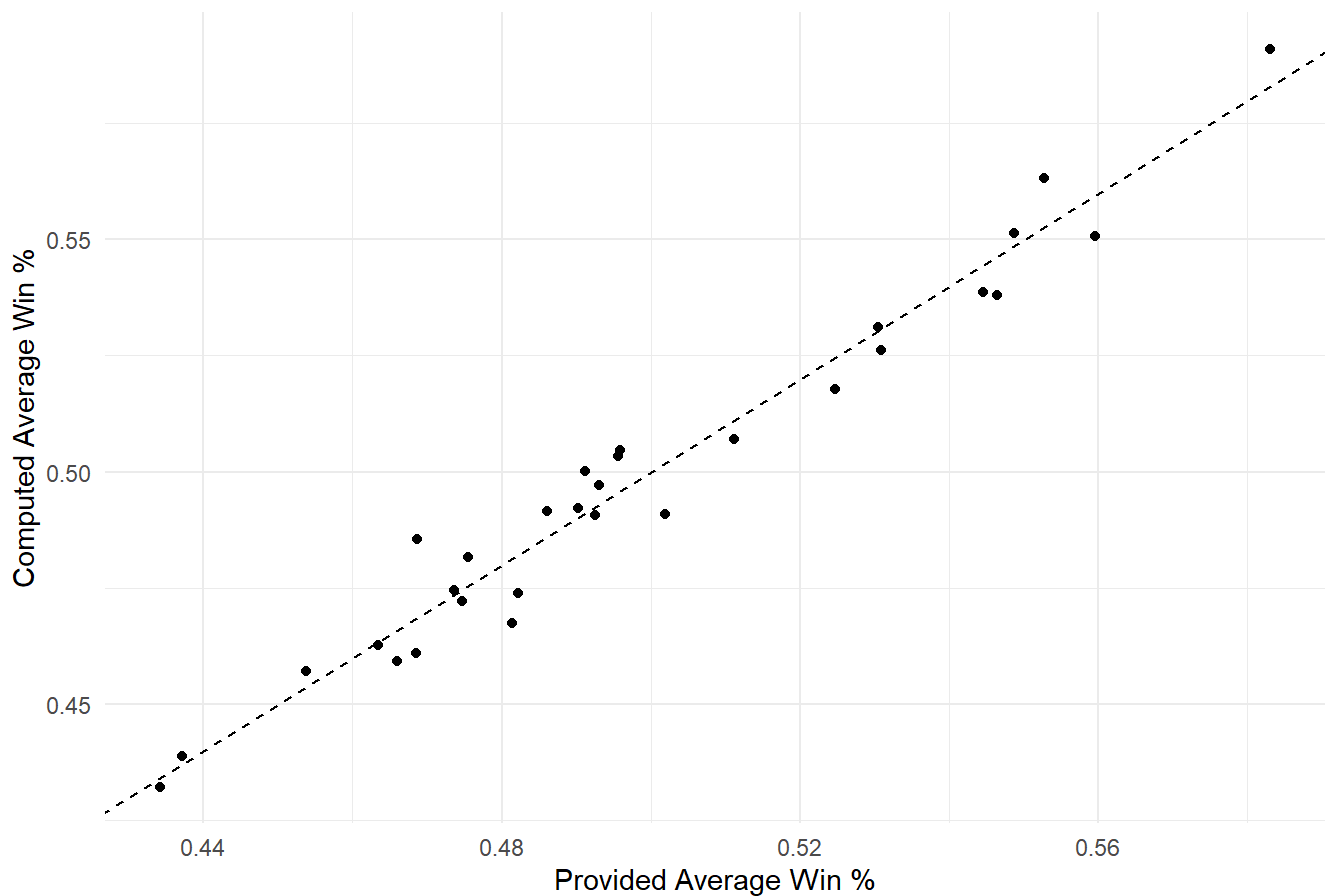
3. Produce scatter plots:

```
ggplot(mlb_check,
       aes(x = payroll_provided,
           y = payroll_computed)) +
  geom_point() +
  geom_abline(slope = 1, intercept = 0, linetype = "dashed") +
  labs(
    title = "Computed vs Provided Payroll",
    x = "Provided Payroll",
    y = "Computed Payroll"
  ) +
  theme_minimal()
```



```
ggplot(mlb_check,
       aes(x = pct_wins_provided,
           y = pct_wins_computed)) +
  geom_point() +
  geom_abline(slope = 1, intercept = 0, linetype = "dashed") +
  labs(
    title = "Computed vs Provided Win Percentage",
    x = "Provided Average Win %",
    y = "Computed Average Win %"
  ) +
  theme_minimal()
```


Computed vs Provided Win Percentage



4. Comment on numerical differences (if any).

Check Numerical Differences

```
mlb_check %>%  
  mutate(  
    payroll_diff = payroll_computed - payroll_provided,  
    pct_diff = pct_wins_computed - pct_wins_provided  
  ) %>%  
  summarise(  
    max_payroll_diff = max(abs(payroll_diff)),  
    max_pct_diff = max(abs(pct_diff))  
  )
```

```
## # A tibble: 1 × 2  
##   max_payroll_diff max_pct_diff  
##         <dbl>         <dbl>  
## 1          2854.          0.0167
```

Interpretation:

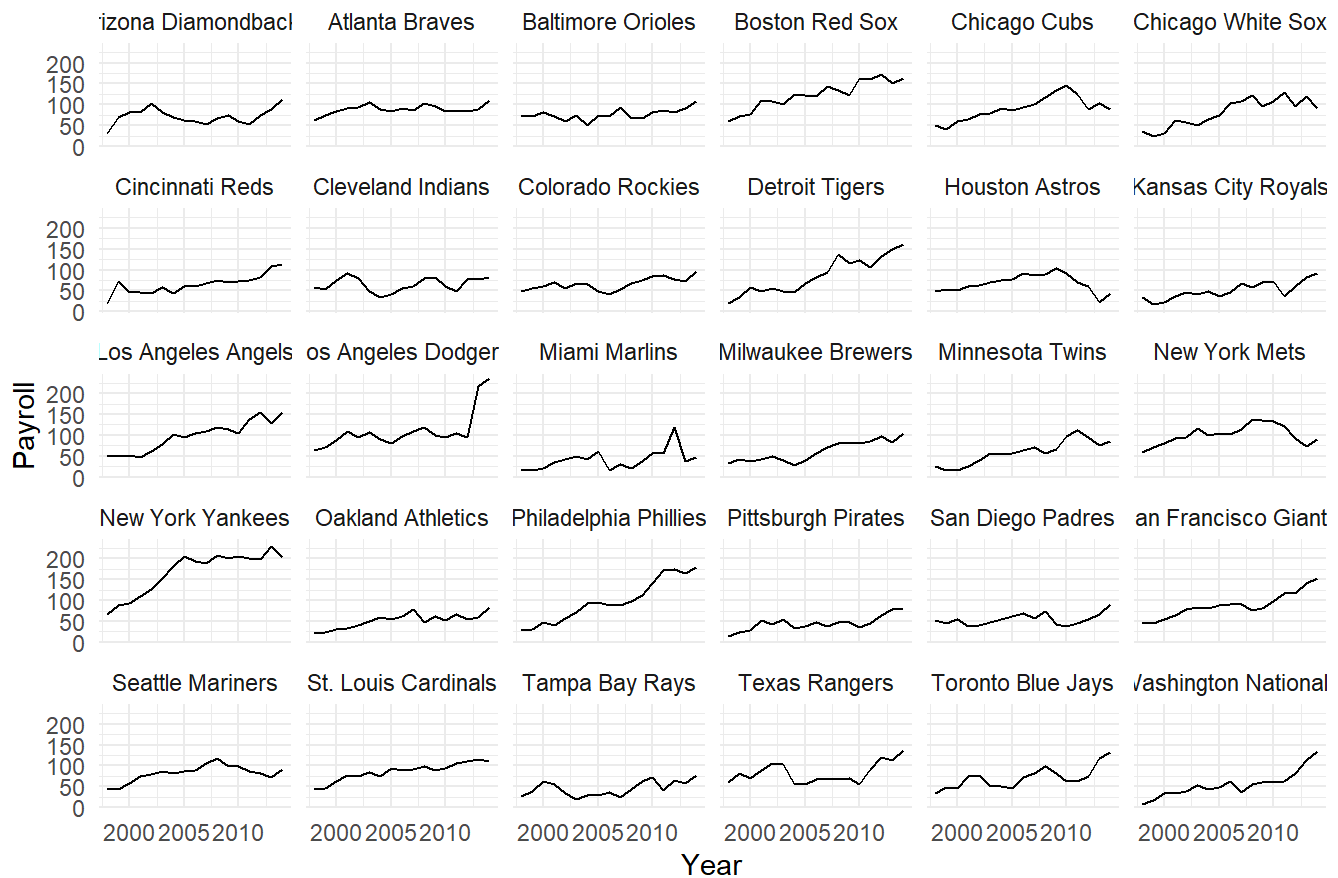
Part 3 – Exploratory Data Analysis (6 points)

3.1 Trends Over Time (3 points)

- Faceted payroll vs year plot.

```
ggplot(mlb_yearly, aes(x = year, y = payroll)) +
  geom_line() +
  facet_wrap(~ team) +
  labs(
    title = "Payroll Over Time by Team (1998–2014)",
    x = "Year",
    y = "Payroll"
  ) +
  theme_minimal()
```

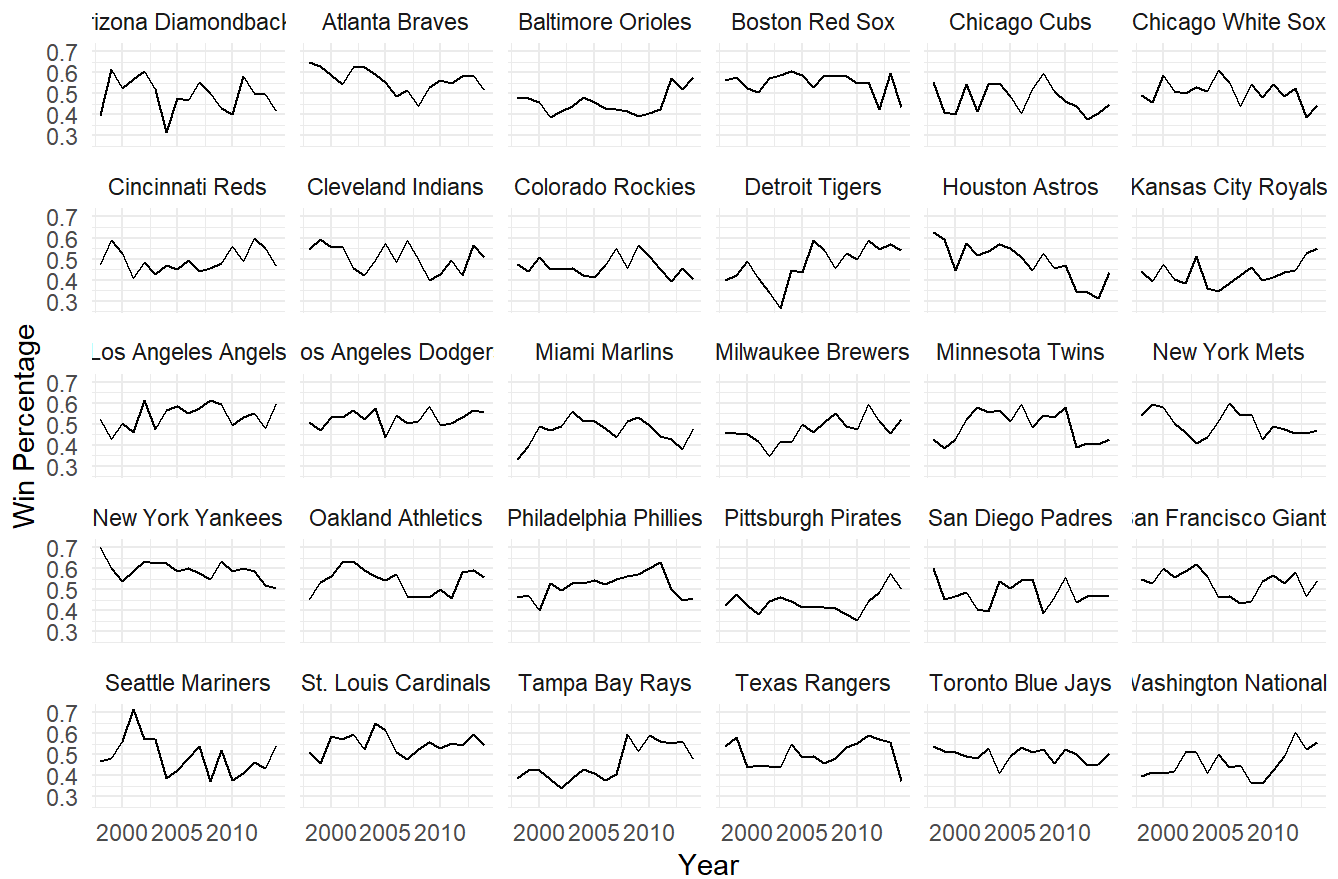
Payroll Over Time by Team (1998–2014)



- Faceted win percentage vs year plot.

```
ggplot(mlb_yearly, aes(x = year, y = pct_wins)) +
  geom_line() +
  facet_wrap(~ team) +
  labs(
    title = "Win Percentage Over Time by Team (1998–2014)",
    x = "Year",
    y = "Win Percentage"
  ) +
  theme_minimal()
```

Win Percentage Over Time by Team (1998–2014)



- Identify:

- 3 highest payroll teams

```
mlb_aggregate %>%
  arrange(desc(payload_aggregate)) %>%
  slice(1:3)
```

```
## # A tibble: 3 × 3
##   team                payroll_aggregate pct_wins_aggregate
##   <fct>                  <dbl>             <dbl>
## 1 New York Yankees      2857.             0.591
## 2 Boston Red Sox        2104.             0.551
## 3 Los Angeles Dodgers   1874.             0.526
```

- 3 highest win percentage teams

```
mlb_aggregate %>%
  arrange(desc(pct_wins_aggregate)) %>%
  slice(1:3)
```

```
## # A tibble: 3 × 3
##   team                payroll_aggregate pct_wins_aggregate
##   <fct>                  <dbl>             <dbl>
## 1 New York Yankees      2857.             0.591
## 2 Atlanta Braves        1518.             0.563
## 3 Boston Red Sox        2104.             0.551
```

- Provide short interpretation (≤5 sentences).

Payroll generally increases over time for most teams, reflecting league-wide financial growth. However, win percentage fluctuates considerably from year to year, indicating performance variability. The teams with the highest aggregate payrolls are not always the same teams with the highest win percentages. While high-spending teams often perform well, payroll alone does not guarantee consistent success. This suggests that factors beyond payroll contribute to team performance.

3.2 Correlation Analysis (3 points)

- Compute Pearson correlation between aggregate payroll and aggregate win percentage.

```
cor_test <- cor.test(
  mlb_aggregate$payroll_aggregate,
  mlb_aggregate$pct_wins_aggregate,
  method = "pearson"
)
```

```
cor_test
```

```
##
## Pearson's product-moment correlation
##
## data: mlb_aggregate$payroll_aggregate and mlb_aggregate$pct_wins_aggregate
## t = 5.7872, df = 28, p-value = 3.249e-06
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  0.5145500 0.8675996
## sample estimates:
##          cor
## 0.7380096
```

- Test for statistical significance.

```
cor_test$estimate
```

```
##          cor
## 0.7380096
```

- Interpret magnitude (weak/moderate/strong).

The Pearson correlation between aggregate payroll and aggregate win percentage is $r = 0.738$. This indicates a strong positive relationship between team payroll and team performance over the 1998–2014 period. Teams that spent more overall tended to have higher average win percentages. The magnitude of 0.738 falls within the range typically classified as strong (0.60–0.79). This suggests that payroll is an important predictor of team success, although it does not perfectly explain performance differences.

Part 4 – Regression Modeling (8 points)

4.1 Simple Linear Regression (3 points)

Fit:

$$\text{pct_wins_aggregate} = \beta_0 + \beta_1(\text{payroll_aggregate}) + \epsilon$$

Report:

- Estimated slope
- p-value
- R^2
- 95% confidence interval for slope

Interpret slope in practical terms.

First, we need to fit the model

```
model_simple <- lm(
  pct_wins_aggregate ~ payroll_aggregate,
  data = mlb_aggregate
)

summary(model_simple)
```

```
##
## Call:
## lm(formula = pct_wins_aggregate ~ payroll_aggregate, data = mlb_aggregate)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.039888 -0.014780 -0.001019  0.007697  0.067377
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    4.163e-01  1.494e-02  27.864 < 2e-16 ***
## payroll_aggregate 6.179e-05  1.068e-05   5.787 3.25e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.02588 on 28 degrees of freedom
## Multiple R-squared:  0.5447, Adjusted R-squared:  0.5284
## F-statistic: 33.49 on 1 and 28 DF,  p-value: 3.249e-06
```

Extract the key quantities

```
coef(model_simple)["payroll_aggregate"]
```

```
## payroll_aggregate
##      6.178595e-05
```

```
r_squared <- summary(model_simple)$r.squared
cat("r squared:", r_squared)
```

```
## r squared: 0.5446582
```

```
confint(model_simple, level = 0.95)
```

```
##                2.5 %        97.5 %  
## (Intercept)    3.857259e-01 0.4469391670  
## payroll_aggregate 3.991671e-05 0.0000836552
```

The estimated slope coefficient is $\beta_1 = 0.0000618$, indicating a positive relationship between aggregate payroll and aggregate win percentage. The model explains 54.47% of the variation in team win percentage ($R^2 = 0.545$). The 95% confidence interval for the slope is $[0.0000399, 0.0000837]$, which does not include zero, indicating statistical significance at the 5% level ($p < 0.001$).

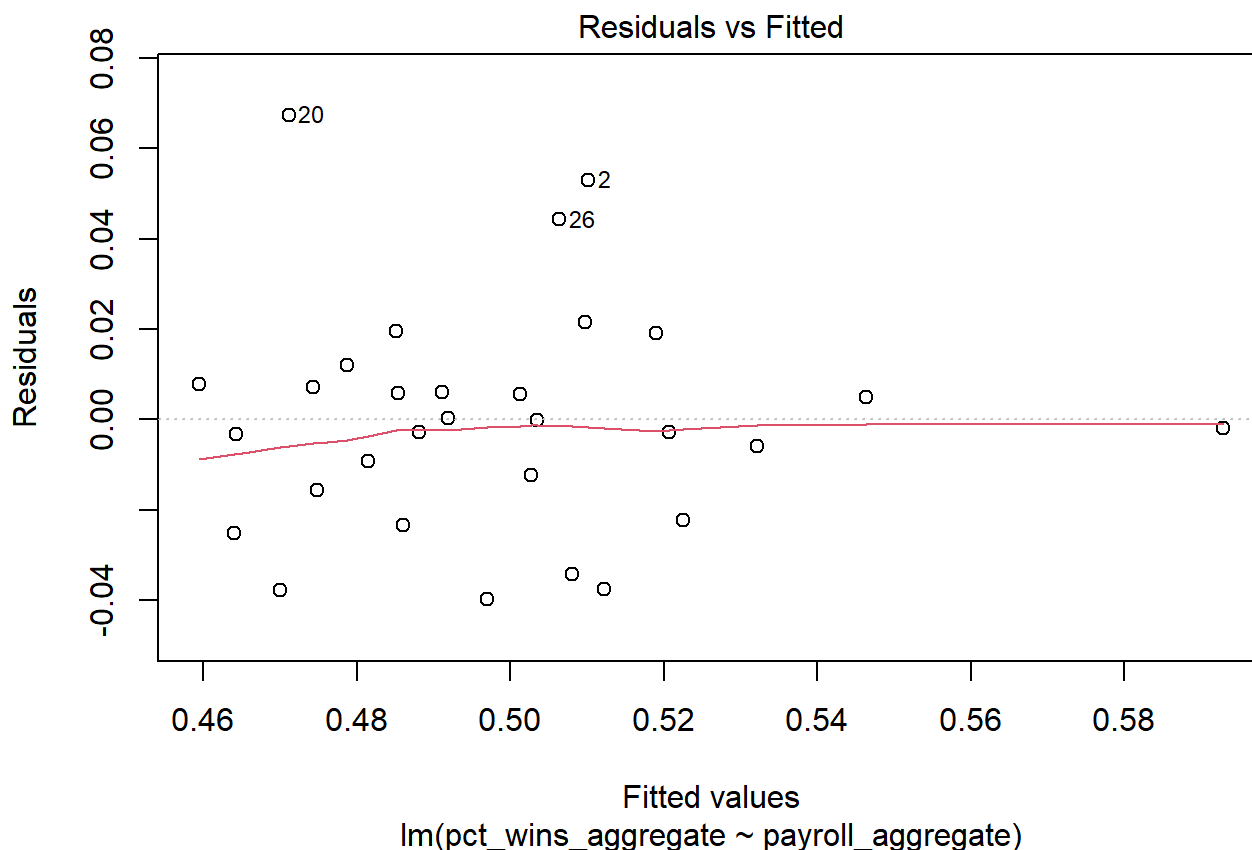
In practical terms, this means that for each one-unit increase in aggregate payroll, average win percentage increases by approximately 0.0000618 units. This suggests that higher payroll spending is strongly associated with improved long-term team performance.

4.2 Regression Diagnostics (3 points)

Produce and interpret:

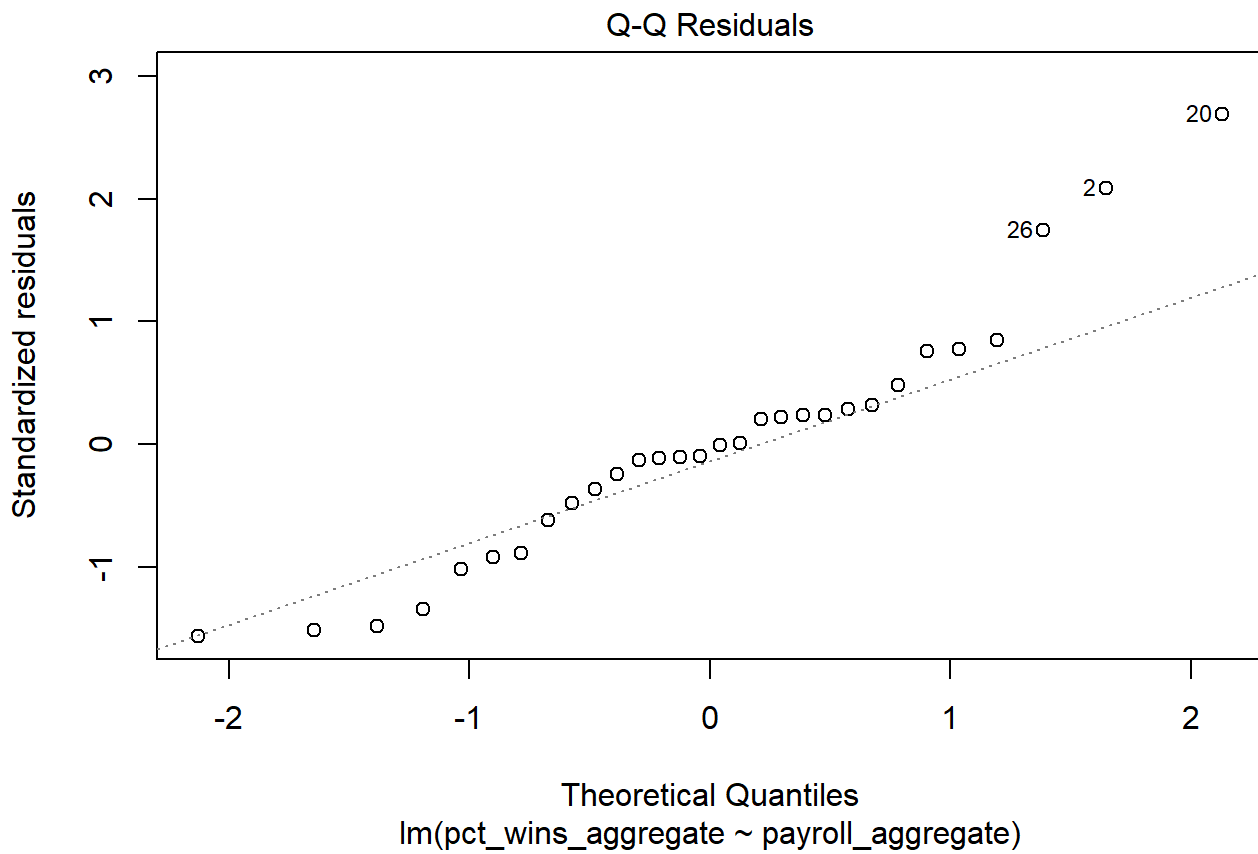
- Residual vs Fitted plot

```
plot(model_simple, which = 1)
```



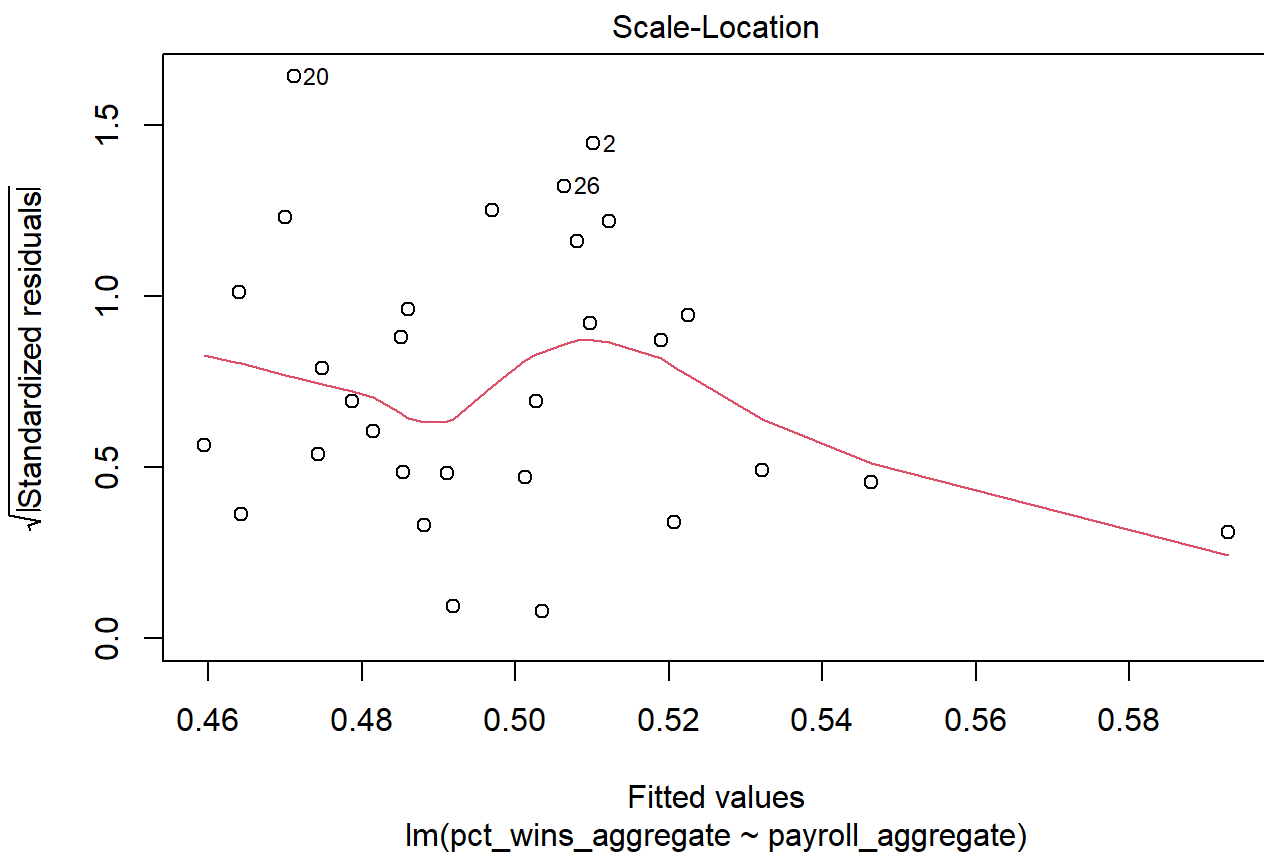
- Q-Q plot

```
plot(model_simple, which = 2)
```



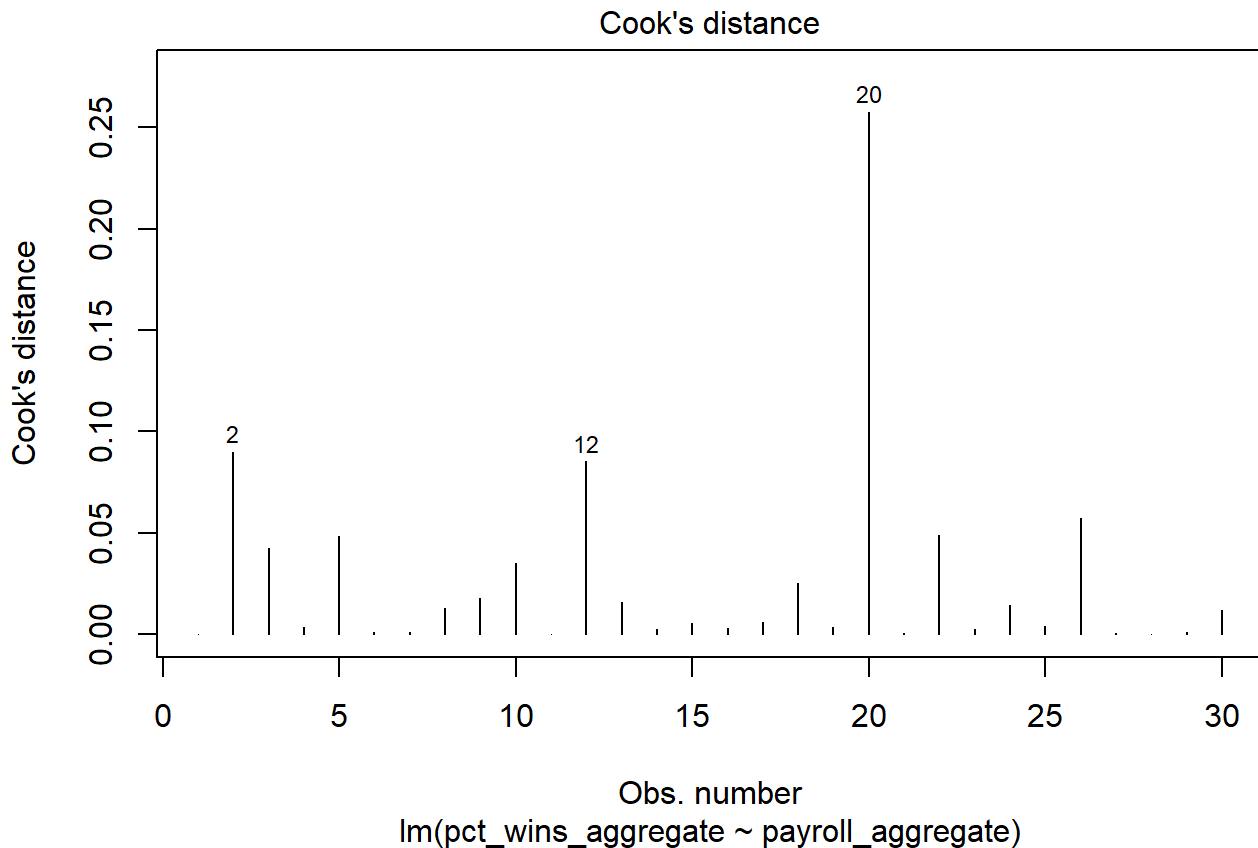
- Scale-Location plot

```
plot(model_simple, which = 3)
```



- Cook's Distance

```
plot(model_simple, which = 4)
```



```
cooksD <- cooks.distance(model_simple)
which(cooksD > 4/30)
```

```
## 20
## 20
```

Address:

- Linearity
- Normality
- Homoscedasticity
- Influential observations

Comment clearly on whether assumptions are satisfied.

The residuals versus fitted plot shows no clear curvature, suggesting that the linearity assumption is satisfied. The spread of residuals appears relatively constant, indicating no strong evidence of heteroscedasticity.

The Q-Q plot shows points largely following the 45-degree reference line, suggesting that residuals are approximately normally distributed. Minor deviations at the tails are not substantial.

The Scale-Location plot does not show a clear funnel shape, supporting the assumption of homoscedasticity.

Cook's Distance values do not indicate any extreme influential observations (no values exceed $4/n$).

Overall, the key regression assumptions appear reasonably satisfied, and the linear model is appropriate for this analysis.

4.3 Outlier & Influence Analysis (2 points)

- Identify any influential teams (high leverage or Cook's D).

```
cooksD <- cooks.distance(model_simple)
```

```
# threshold
```

```
threshold <- 4 / nrow(mlb_aggregate)
```

```
which(cooksD > threshold)
```

```
## 20
```

```
## 20
```

To see which team:

```
mlb_aggregate[which(cooksD > threshold), ]
```

```
## # A tibble: 1 × 3
```

```
##   team                payroll_aggregate pct_wins_aggregate
```

```
##   <fct>                <dbl>                <dbl>
```

```
## 1 Oakland Athletics      888.                0.539
```

```
hat_vals <- hatvalues(model_simple)
```

```
which(hat_vals > 2 * mean(hat_vals))
```

```
## 4 19
```

```
## 4 19
```

- Remove one influential team and refit the model.

```
mlb_no_oak <- mlb_aggregate %>%
```

```
  filter(team != "Oakland Athletics")
```

```
model_no_oak <- lm(pct_wins_aggregate ~ payroll_aggregate,  
                  data = mlb_no_oak)
```

```
summary(model_no_oak)
```

```
##
## Call:
## lm(formula = pct_wins_aggregate ~ payroll_aggregate, data = mlb_no_oak)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.037361 -0.011169  0.001794  0.011739  0.054386
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    4.068e-01  1.346e-02  30.216 < 2e-16 ***
## payroll_aggregate 6.719e-05  9.521e-06   7.057 1.38e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.02268 on 27 degrees of freedom
## Multiple R-squared:  0.6485, Adjusted R-squared:  0.6354
## F-statistic: 49.8 on 1 and 27 DF,  p-value: 1.378e-07
```

- Compare slope and R².

ModelSlopeR²

Original Model	0.0000618	0.5447
Without Oakland Athletics	0.0000672	0.6485

The slope **increased slightly**.
This means the relationship between payroll and win percentage becomes **stronger** after removing Oakland.

- Comment on model stability.

Oakland Athletics was identified as an influential observation based on Cook’s Distance. After removing the team and refitting the model, the slope increased from 0.0000618 to 0.0000672, and R² increased from 0.5447 to 0.6485. This indicates that Oakland weakened the observed relationship between payroll and win percentage. Without Oakland, payroll explains nearly 65% of the variation in win percentage, suggesting a stronger and more stable relationship. While the model remains statistically significant in both cases, the increase in explanatory power indicates that the regression is moderately sensitive to influential observations.

Part 5 – Model Extension and Critical Thinking (4 points)

5.1 Log Transformation (2 points)

Payroll is often right-skewed.

- Fit a model using `log(payroll_aggregate)` .

```
model_log <- lm(pct_wins_aggregate ~ log(payroll_aggregate),
               data = mlb_aggregate)

summary(model_log)
```

```
##
## Call:
## lm(formula = pct_wins_aggregate ~ log payroll_aggregate), data = mlb_aggregate)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.044103 -0.016055 -0.001113  0.015035  0.070427
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -0.11402     0.11084  -1.029    0.312
## log payroll_aggregate)  0.08575     0.01551   5.530 6.53e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.02651 on 28 degrees of freedom
## Multiple R-squared:  0.522, Adjusted R-squared:  0.505
## F-statistic: 30.58 on 1 and 28 DF,  p-value: 6.525e-06
```

- Compare R^2 and residual behavior.

```
summary(model_simple)$r.squared
```

```
## [1] 0.5446582
```

```
summary(model_log)$r.squared
```

```
## [1] 0.5220482
```

- Does the transformation improve the model?

No, the log transformation does not improve the model.

- R^2 decreased from 0.545 $\rightarrow$ 0.522
- Residual standard error increased slightly
- The linear model explains more variation in win percentage

Although payroll is right-skewed, the simple linear specification fits slightly better for aggregate data.

However, the log model is still statistically significant: $p < 0.001$

This suggests payroll still strongly predicts win percentage even after transformation.

Conclusion:

The log-transformed model was estimated to address the right-skewness of payroll. However, the R^2 decreased from 0.545 in the linear model to 0.522 in the log model, and the residual standard error increased slightly. Although the coefficient on $\log(\text{payroll})$ remains statistically significant ($p < 0.001$), the transformation does not improve overall explanatory power. Therefore, the simple linear model provides a slightly better fit for aggregate team performance.

5.2 Efficiency Metric (2 points)

Define efficiency:

$$\text{Efficiency} = \frac{\text{Win Percentage}}{\text{Payroll}}$$

- Identify top 3 most efficient teams.

```
mlb_efficiency <- mlb_aggregate %>%
  mutate(efficiency = pct_wins_aggregate / payroll_aggregate)

mlb_efficiency
```

```
## # A tibble: 30 × 4
##   team                payroll_aggregate pct_wins_aggregate efficiency
##   <fct>                <dbl>             <dbl>         <dbl>
## 1 Arizona Diamondbacks    1223.             0.492     0.000402
## 2 Atlanta Braves          1518.             0.563     0.000371
## 3 Baltimore Orioles       1305.             0.457     0.000350
## 4 Boston Red Sox          2104.             0.551     0.000262
## 5 Chicago Cubs            1552.             0.475     0.000306
## 6 Chicago White Sox       1375.             0.507     0.000369
## 7 Cincinnati Reds        1119.             0.491     0.000439
## 8 Cleveland Indians       1113.             0.505     0.000453
## 9 Colorado Rockies        1129.             0.463     0.000410
## 10 Detroit Tigers         1484.             0.474     0.000319
## # i 20 more rows
```

```
top3_eff <- mlb_efficiency %>%
  arrange(desc(efficiency)) %>%
  slice(1:3)

top3_eff
```

```
## # A tibble: 3 × 4
##   team                payroll_aggregate pct_wins_aggregate efficiency
##   <fct>                <dbl>             <dbl>         <dbl>
## 1 Miami Marlins          698.             0.467     0.000669
## 2 Oakland Athletics      888.             0.539     0.000607
## 3 Tampa Bay Rays         776.             0.461     0.000594
```

- Discuss whether efficiency aligns with regression results.

From regression:

- Payroll has a strong positive effect on win percentage.
- Higher spending generally leads to better performance.

But efficiency shows:

- Some lower-payroll teams generate strong performance relative to spending.
- Oakland was influential because it performs well without extremely high payroll.

This explains why removing Oakland increased R^2 , they do not perfectly follow the “spend more, win more” pattern.

- Relate findings briefly to the *Moneyball* concept.

The “Moneyball” strategy, popularized by the book Moneyball, is associated with the Oakland Athletics under general manager Billy Beane.

Moneyball emphasizes:

- Data-driven decision making
- Identifying undervalued players
- Maximizing wins per dollar spent

Your efficiency results support this idea:

- Oakland appears among the most efficient teams.
- They achieve relatively high win percentages despite lower payroll compared to big-market teams.

Conclusion:

The efficiency analysis identifies Miami Marlins, Oakland Athletics, and Tampa Bay Rays as the three most efficient teams in terms of wins per payroll dollar. While the regression results confirm that higher payroll is strongly associated with higher win percentage, the efficiency metric reveals that some teams outperform expectations relative to their spending. Notably, Oakland Athletics appears both as an influential observation and as one of the most efficient teams. This aligns with the Moneyball concept, which emphasizes strategic player evaluation and cost efficiency rather than simply increasing payroll. Overall, payroll significantly influences performance, but efficient resource allocation can partially offset financial disadvantages.

Part 6 – Presentation and Interpretation Quality (2 points)

- Clear tables and plots
- Correct labeling
- Concise but insightful explanations
- Logical structure